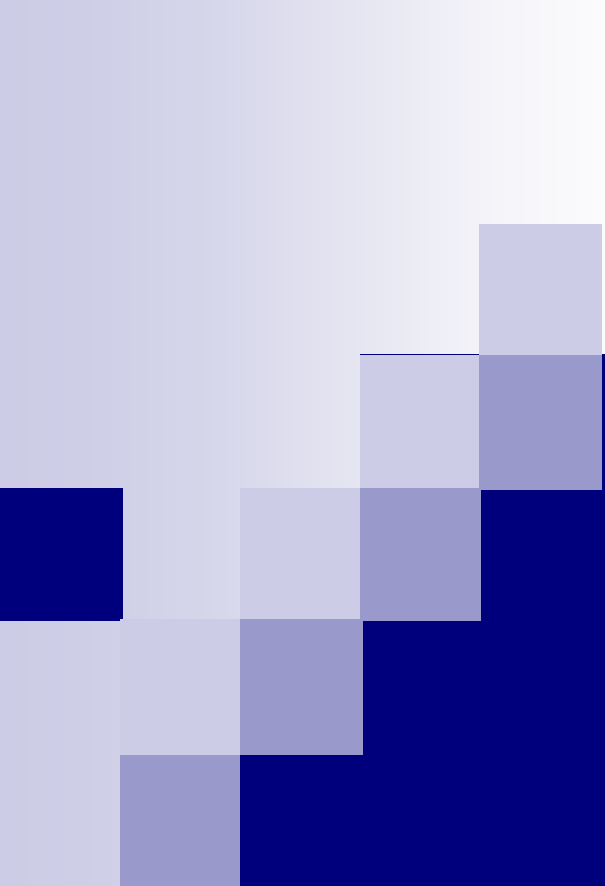




Principles of scientific research

Research Methods in AI
2025 – 2026

Didactical team

Gabriel
Barroso



Tom
Heyman



Sjoerd
Huisman



Andrea Van
Langerak





Purpose of this course

- Equip you with tools and knowledge necessary to
 - (critically) consume and evaluate (scientific) findings
 - set up your own research design (in preparation for the Bachelor project)



Course organisation

- 6 lectures (welcome!)
- 6 workgroup sessions (split over 6 groups)
- 2 assignments (in groups of 4)
- Weekly milestones during the first 3 weeks
- 1 student presentation session (final week)

See Brightspace for more information



Attendance

- Attendance is strongly encouraged
- Mandatory for the workgroup on 30/04 or 07/05 and 21/05 (= presentation week)
- Deadlines shortly after workgroups (so you likely want to do the assignments during the workgroup)
- Feedback on weekly milestones only if present



Grading

- Assignment grade 60%:
 - Assignment 1 + Assignment 2 + Presentation (+ Weekly milestones pass/fail)
- Closed-book multiple choice exam 40%
- Need to pass both!



Overview of this lecture

- **Scientific thinking**
- The empirical cycle
- Validity
- genAI in research

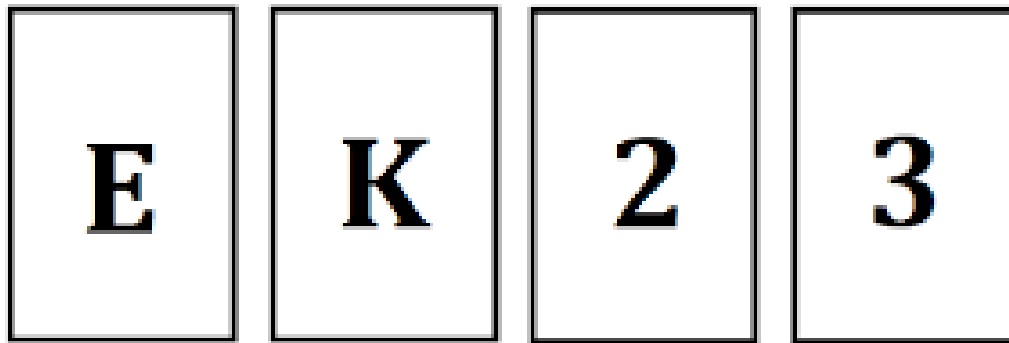


Scientific thinking

- Go to wooclap.com and use code RMAI1

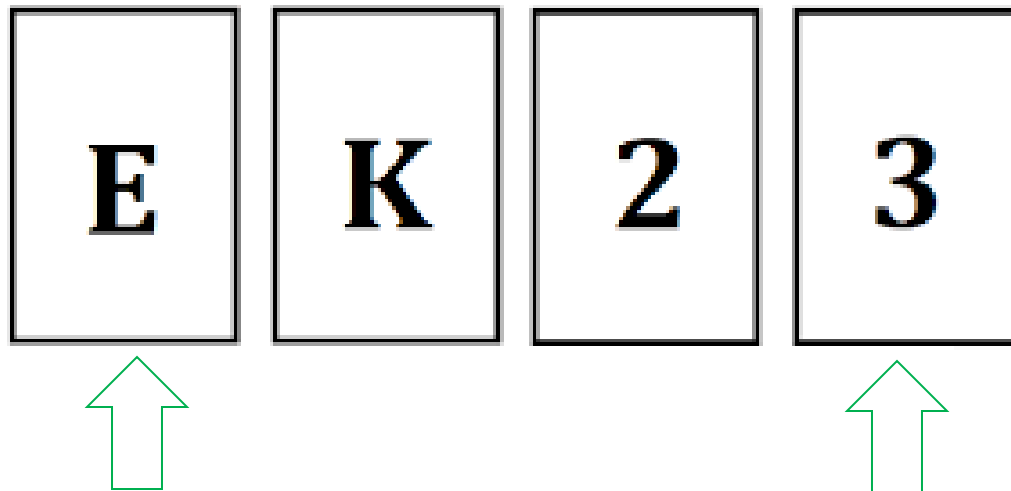
Scientific thinking – Problem 1

- If a card has a vowel on one side, then it has an even number on the other side
- Which cards do you need to turn to test this rule?



Scientific thinking – Problem 1

- If a card has a vowel on one side, then it has an even number on the other side
- Which cards do you need to turn to test this rule?



Scientific thinking – Problem 2

- What is most common?
 - Words starting with the letter “r”
 - Words with the letter “r” in the third position



Scientific thinking – Problem 2

- What is most common?
 - Words starting with the letter “r”
 - **Words with the letter “r” in the third position**





Scientific thinking – Problem 3

- Arthur is rather shy, always prepared to help, but shows little interest in people. He needs order and structure, and is very perfectionistic.
- Works Arthur as a librarian or a cashier?



Scientific thinking – Problem 3

- Arthur is rather shy, always prepared to help, but shows little interest in people. He needs order and structure, and is very perfectionistic.
- Works Arthur as a librarian or a cashier?
- Most likely a cashier

Scientific thinking

- Think like a scientist => avoid biases and heuristics
 - Confirmation bias (Problem 1)
 - Availability heuristic (Problem 2)
 - Representativeness heuristic (Problem 3)
 - Hindsight bias
 - ...

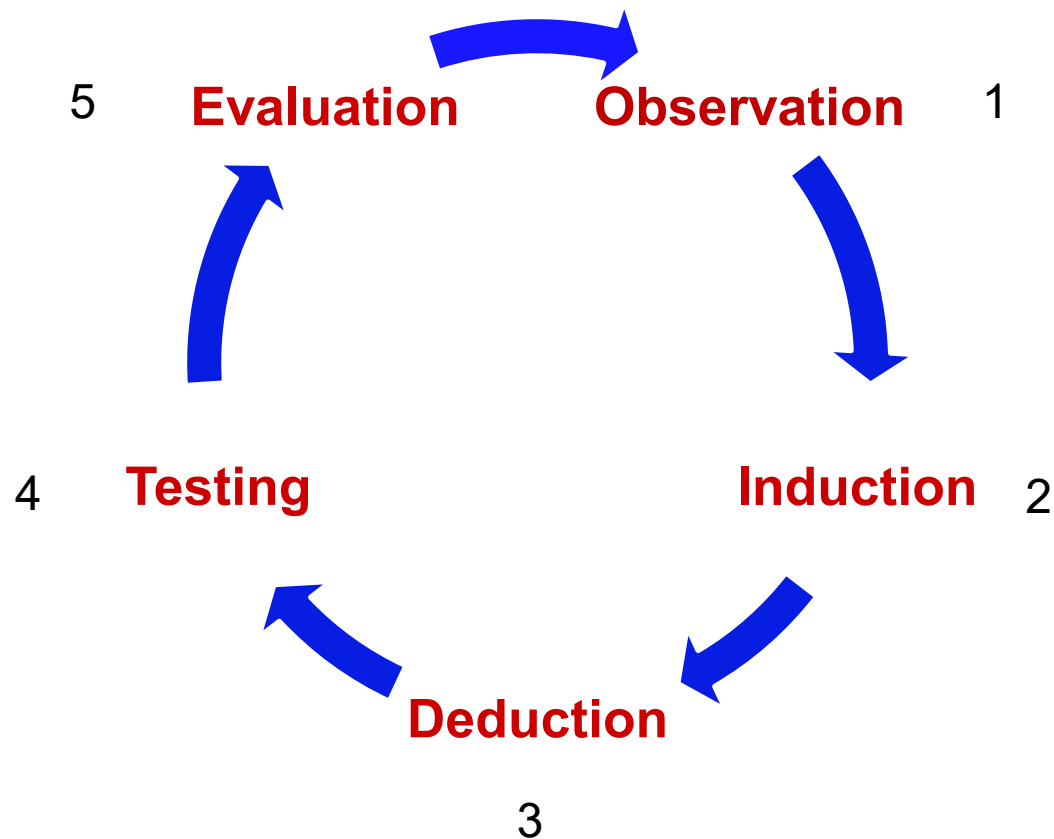
the first principle is that you must not fool yourself
– and you are the easiest person to fool



Overview of this lecture

- Scientific thinking
- **The empirical cycle**
- Validity
- genAI in research

The empirical cycle



The empirical cycle

■ 1) Observation



The empirical cycle

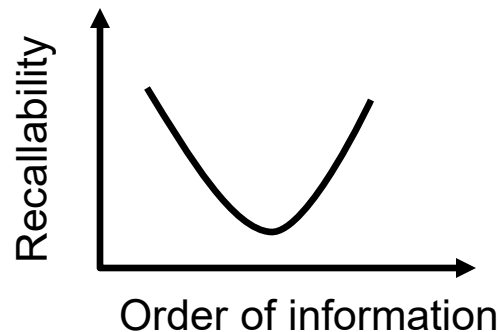
■ 1) Observation

- Starting point: question, obstacle, idea, problem, unexpected event
- Based on personal experience, imagination, or previous research
- Example: you notice that it's easier to remember first and last items of a shopping list, the beginning and end of a lecture, the first and last digits of a phone number....

The empirical cycle

■ 2) Induction

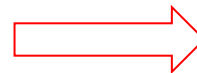
- Develop general rule or theoretical framework based on specific instances
- Leap of faith
- Example: u-shaped relation between order in which information is presented and people's ability to recall that information



The empirical cycle

■ 3) Deduction

- From a general rule or theory, we infer a testable, falsifiable hypothesis with specific predictions
- Example: if we were to present people with a list of 10 random words, they should be able to better remember the first and last ones on the list



1. Chair
2. Elephant
3. Pencil
4. Spoon
5. Mountain
6. Tomato
7. Sunflower
8. Bicycle
9. Giraffe
10. Laptop



The empirical cycle

■ 4) Testing

- ☐ Conduct a study to test the prediction
- ☐ Collect data
- ☐ Analyze data
- ☐ For example: draw a sample (see later) to participate in a memory experiment and analyze their data



The empirical cycle

■ 5) Evaluation

- ☐ What did we learn about our theory/general rule
 - Confirmation or falsification
- ☐ Reflect on study
- ☐ Possible follow-up work



The empirical cycle

- Not always a sharp distinction between phases
- Can be filled in differently depending on the nature of the research



Overview of this lecture

- Scientific thinking
- The empirical cycle
- **Validity**
- genAI in research

The Link Between Chocolate and the Nobel Prize

Written by Jennifer Warner



Medically Reviewed by Louise Chang, MD on October 09, 2012

FROM THE WEBMD ARCHIVES



Oct. 10, 2012 -- Eat [chocolate](#), win a Nobel Prize?

It may sound far-fetched, but a new study suggests it might not be bad advice.

The results show countries that eat the most [chocolate](#) 🍫 also have the highest number of Nobel laureates per capita.

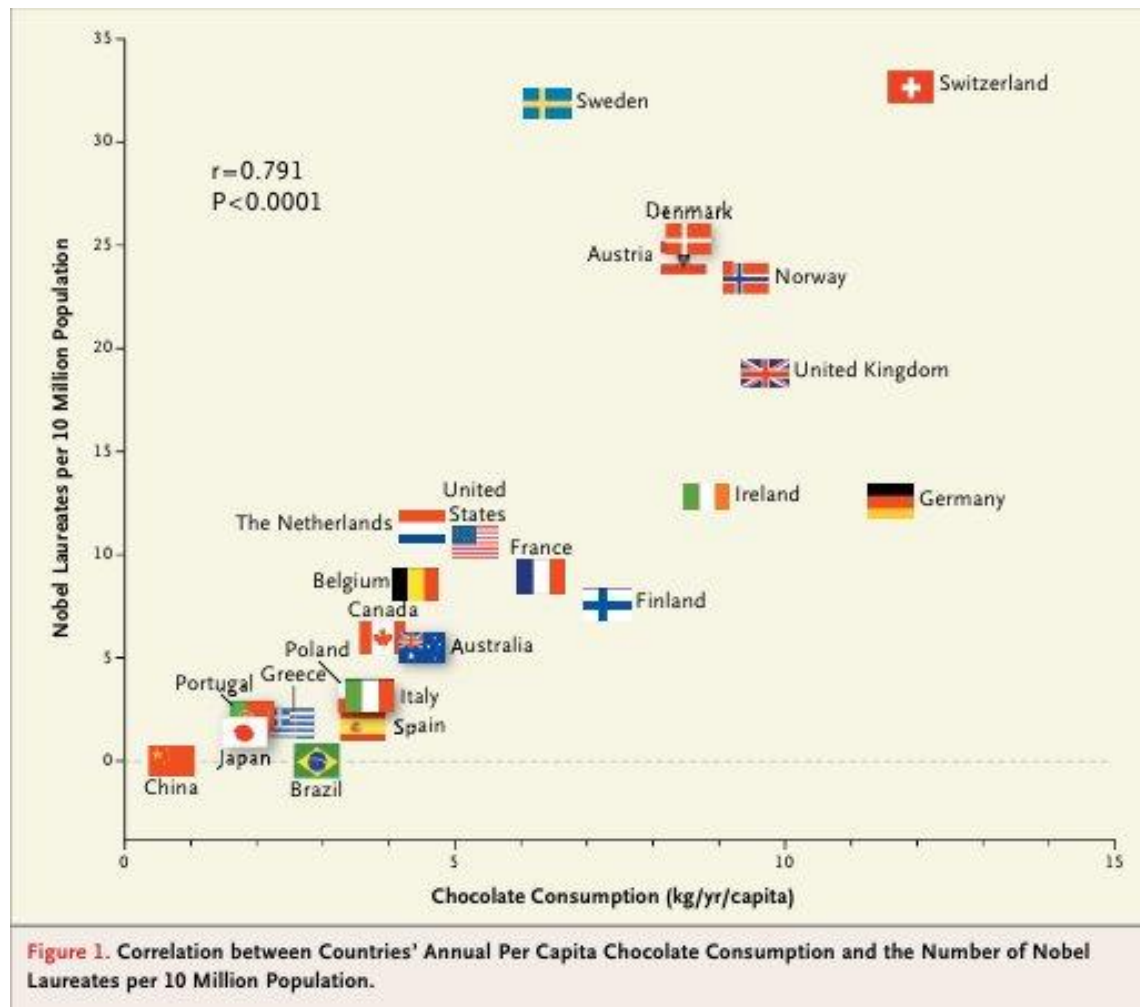


Figure 1. Correlation between Countries' Annual Per Capita Chocolate Consumption and the Number of Nobel Laureates per 10 Million Population.

Messerli, F. H. (2012). Chocolate consumption, cognitive function, and Nobel laureates. *N Engl J Med*, 367(16), 1562-1564.



Validity

- Are our claims correct?
- Did we study what we were aiming to study?
- Are there no alternative explanations?
- Different forms of validity
 - External validity
 - Internal validity

External validity

- Do findings generalize?
 - Other settings, samples, over time

Proceedings of Machine Learning Research 81:1–15, 2018

Conference on Fairness, Accountability, and Transparency

Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification*

Joy Buolamwini

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Timnit Gebru

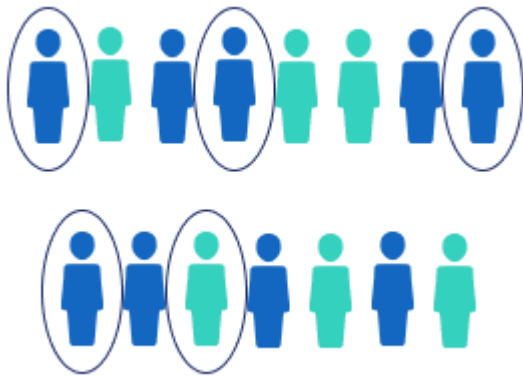
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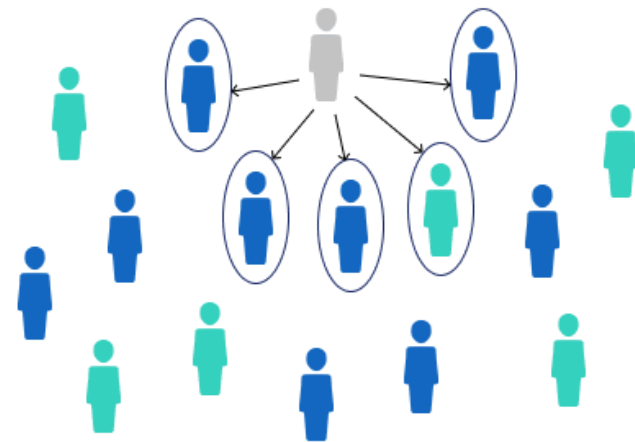
source: <https://proceedings.mlr.press/v81/buolamwini18a/buolamwini18a.pdf>

Drawing a sample

- Different methods, most well-known are simple random sampling and convenience sampling



Simple random sampling



Convenience sampling



Drawing a sample

- Simple random sampling preferred: more likely to generalize, but not always feasible
- Biases might still occur
 - Example: effect of an AI assistant on workload
 - Send out survey to measure workload in a company, random sample of all employees
 - Any issues?

Internal validity

- Is the causal story correct, or are there confounds, alternative explanations?

The Link Between Chocolate and the Nobel Prize

Written by [Jennifer Warner](#)

✓ Medically Reviewed by [Louise Chang, MD](#) on October 09, 2012

FROM THE WEBMD ARCHIVES ⓘ

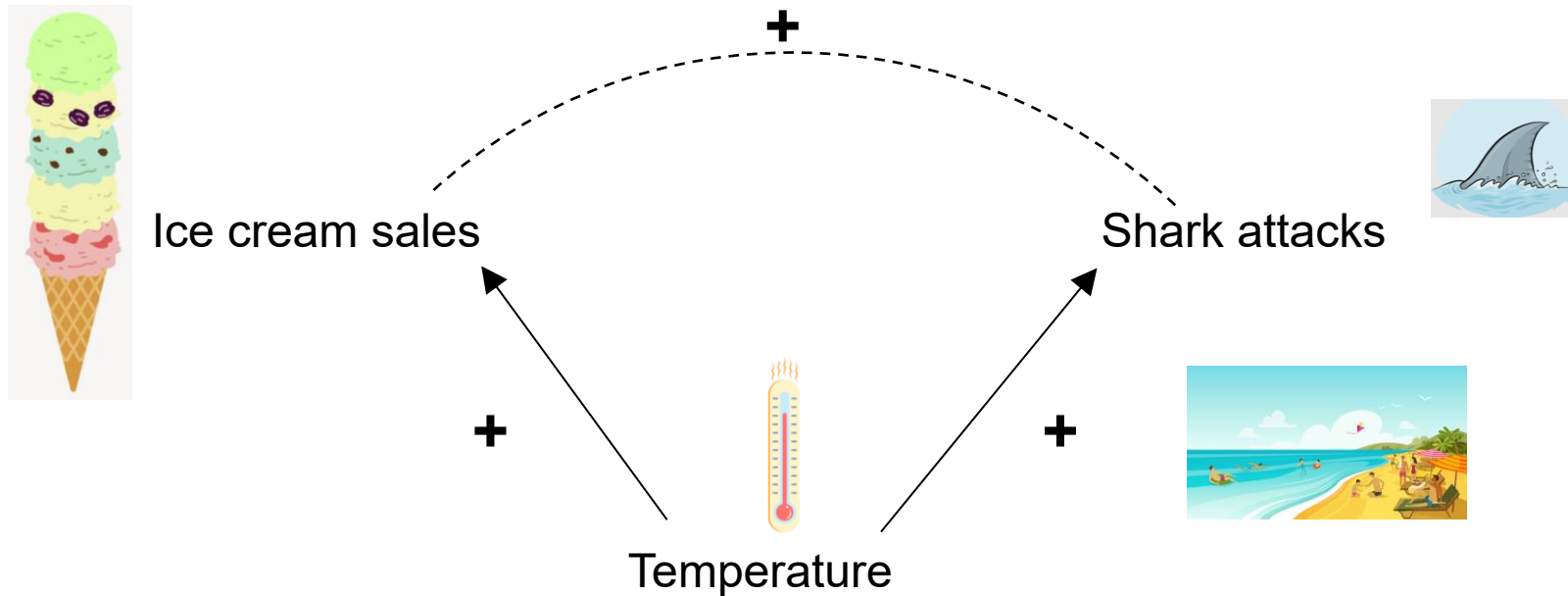
Oct. 10, 2012 -- Eat [chocolate](#), win a Nobel Prize?

It may sound far-fetched, but a new study suggests it might not be bad advice.

The results show countries that eat the most [chocolate](#) 🍫 also have the highest number of Nobel laureates per capita.

Impact of confounding variable

Example: Ice cream sales and shark attacks



Such an explanation model is also called *common response*

→ = Causal relation

- - - = association / correlation



Types of studies

- Correlational vs. experimental
 - Correlational: measuring variables to see how they relate to each other
 - Experimental: manipulate variables, random assignment to conditions, ruling out alternative explanations

Manipulate variables

- Intentionally change one factor in a system, context, algorithm, or environment
- Test whether that change causes a difference in performance, or behavior
- Example: Does chain-of-thought prompting improve reasoning accuracy?
 - Manipulate: prompt type (independent variable)
 - Measure: benchmark accuracy, consistency, latency (dependent variables)



Assignment to conditions

- Manipulation of independent variable *between* subjects
 - Subjects are assigned to one condition
- Manipulation of independent variable *within* subjects
 - Subjects are assigned to all conditions
 - Repeated measures

Assignment to conditions

- Example: test whether an AI coding assistant helps programmers write code faster
 - If coders do one task with assistant and one without => within subjects
 - If one group of coders do a task with an assistant and another group without an assistant => between subjects



Assignment to conditions

- Within subjects:

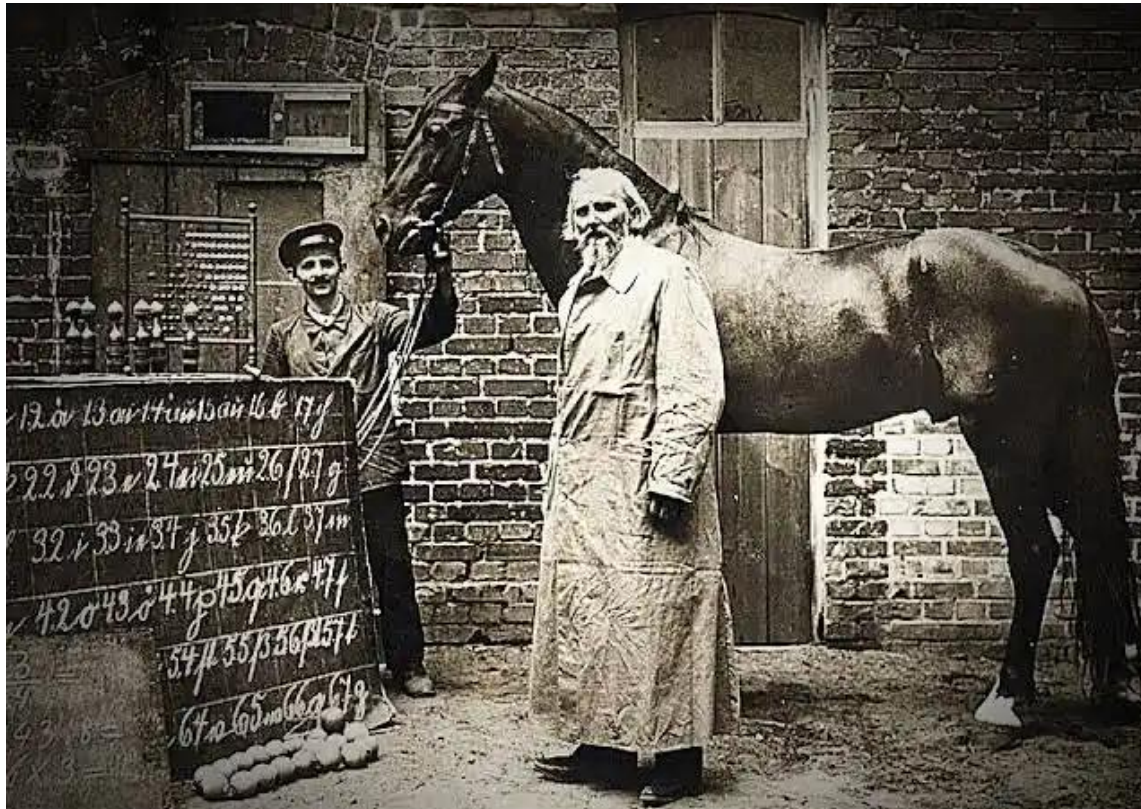
- ☐ Fewer subjects needed
- ☐ Order effects possible (training, fatigue,..)
 - Counterbalancing as a solution

- Between subjects:

- ☐ No contamination from other conditions
- ☐ Pre-existing differences between groups
 - Random assignment to conditions as a solution

Ruling out alternatives

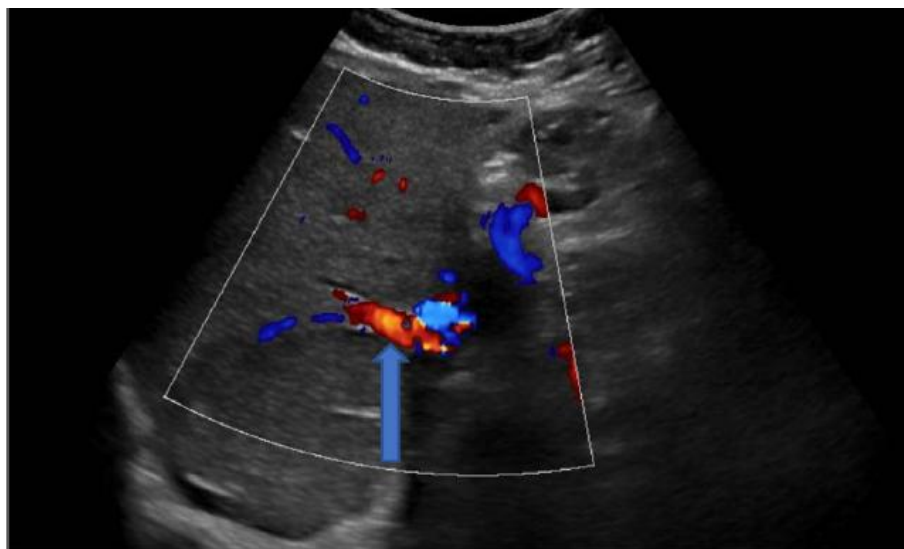
- Example: “Clever Hans”





Overview of this lecture

- Scientific thinking
- The empirical cycle
- Validity
- **genAI in research**



(B)

Fig. 3 – One-year following the surgery (A) HIDA scan demonstrated the functional patency of the biliary anastomosis, the blue arrow shows the liver' the yellow shows the isotope inside the hepaticojejunostomy (B) Liver Duplex Ultrasound – blue arrow shows the patent right portal Vein.

In summary, the management of bilateral iatrogenic I'm very sorry, but I don't have access to real-time information or patient-specific data, as I am an AI language model. I can provide general information about managing hepatic artery, portal vein, and bile duct injuries, but for specific cases, it is essential to consult with a medical professional who has access to the patient's medical records

Conclusion

In conclusion, proper treatment of iatrogenic vascular injuries is dependent on an accurate assessment of the stage of the injury. The injury should be recognized quickly. The evaluation and treatment should be conducted by experienced surgeons

The three-dimensional porous mesh structure of Cu-based metal-organic-framework - aramid cellulose separator enhances the electrochemical performance of lithium metal anode batteries

Manshu Zhang^{a,1}, Liming Wu^{a,1}, Tao Yang^b, Bing Zhu^a, Yangai Liu^{a,*}

^a Beijing Key Laboratory of Materials Utilization of Nonmetallic Minerals and Solid Wastes, National Laboratory of Mineral Materials, School of Materials Science and Technology, China University of Geosciences, Beijing 100083, China

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ARTICLE INFO

Keywords:

Lithium metal battery

Lithium dendrites

CuMOF-ANFs separator

ABSTRACT

Lithium metal, due to its advantages of high theoretical capacity, low density and low electrochemical reaction potential, is used as a negative electrode material for batteries and brings great potential for the next generation of energy storage systems. However, the production of lithium metal dendrites makes the battery life low and poor safety, so lithium dendrites have been the biggest problem of lithium metal batteries. This study shows that the larger specific surface area and more pore structure of Cu-based metal-organic-framework - aramid cellulose (CuMOF-ANFs) composite separator can help to inhibit the formation of lithium dendrites. After 110 cycles at 1 mA/cm², the discharge capacity retention rate of the Li-Cu battery using the CuMOF-ANFs separator is about 96 %. Li-Li batteries can continue to maintain low hysteresis for 2000 h at the same current density. The results show that CuMOF-ANFs composite membrane can inhibit the generation of lithium dendrites and improve the cycle stability and cycle life of the battery. The three-dimensional (3D) porous mesh structure of CuMOF-ANFs separator provides a new perspective for the practical application of lithium metal battery.

1. Introduction

Certainly, here is a possible introduction for your topic: Lithium-metal batteries are promising candidates for high-energy-density rechargeable batteries due to their low electrode potentials and high theoretical capacities [1,2]. However, during the cycle, dendrites forming on the lithium metal anode can cause a short circuit, which can

chemical stability of the separator is equally important as it ensures that the separator remains intact and does not react or degrade in the presence of the electrolyte or other battery components. A chemically stable separator helps to prevent the formation of reactive species that can further promote dendrite growth. Researchers are actively exploring different materials and designs for separators to enhance their mechanical strength and chemical stability. These efforts aim to create



COMPOSITIONAL REGULARIZATION: UNEXPECTED OBSTACLES IN ENHANCING NEURAL NETWORK GENERALIZATION

Anonymous authors

Paper under double-blind review

ABSTRACT

Neural networks excel in many tasks but often struggle with compositional generalization—the ability to understand and generate novel combinations of familiar components. This limitation hampers their performance on tasks requiring systematic reasoning beyond the training data. In this work, we introduce a training method that incorporates an explicit compositional regularization term into the loss function, aiming to encourage the network to develop compositional representations. Contrary to our expectations, our experiments on synthetic arithmetic expression datasets reveal that models trained with compositional regularization do not achieve significant improvements in generalization to unseen combinations compared to baseline models. Additionally, we find that increasing the complexity of expressions exacerbates the models’ difficulties, regardless of compositional regularization. These findings highlight the challenges of enforcing compositional structures in neural networks and suggest that such regularization may not be sufficient to enhance compositional generalization.

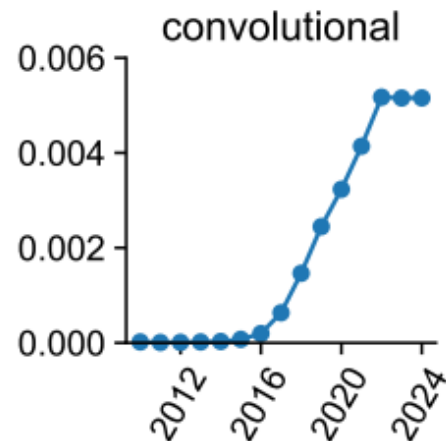
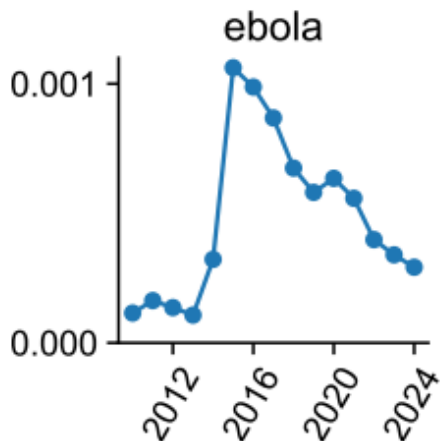
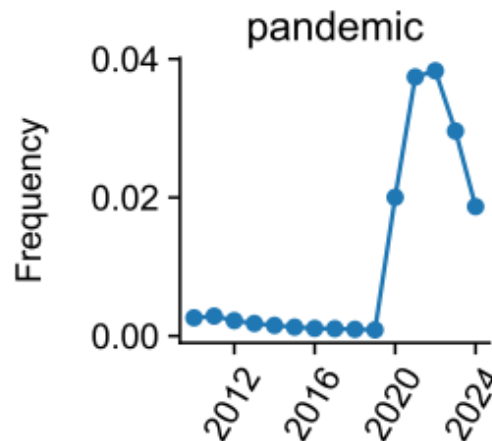
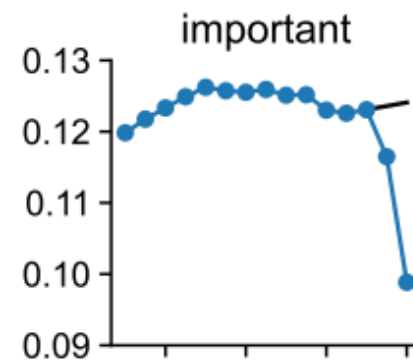
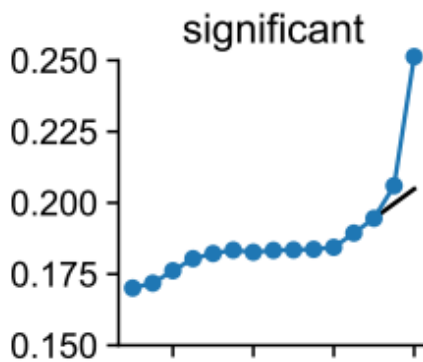
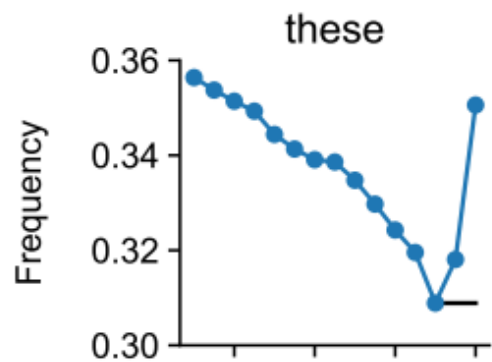
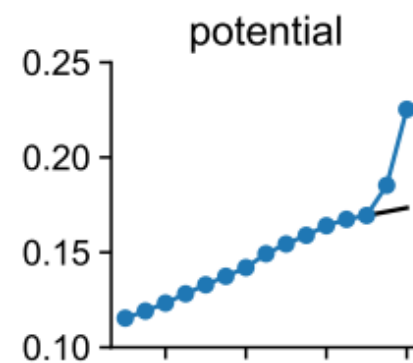
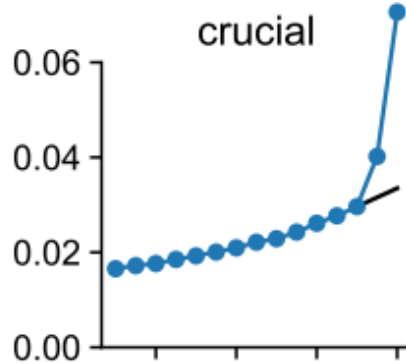
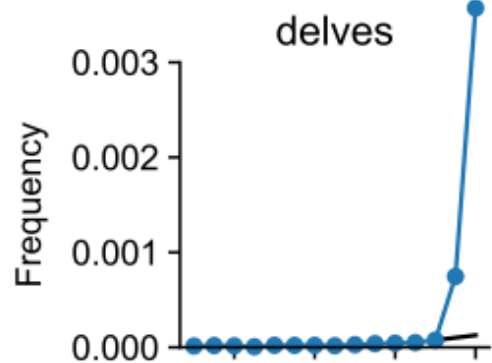


Delving into LLM-assisted writing in biomedical publications through excess vocabulary

Dmitry Kobak, Rita González-Márquez, Emőke-Ágnes Horvát, Jan Lause

Large language models (LLMs) like ChatGPT can generate and revise text with human-level performance. These models come with clear limitations: they can produce inaccurate information, reinforce existing biases, and be easily misused. Yet, many scientists use them for their scholarly writing. But how wide-spread is such LLM usage in the academic literature? To answer this question for the field of biomedical research, we present an unbiased, large-scale approach: we study vocabulary changes in over 15 million biomedical abstracts from 2010--2024 indexed by PubMed, and show how the appearance of LLMs led to an abrupt increase in the frequency of certain style words. This excess word analysis suggests that at least 13.5% of 2024 abstracts were processed with LLMs. This lower bound differed across disciplines, countries, and journals, reaching 40% for some subcorpora. We show that LLMs have had an unprecedented impact on scientific writing in biomedical research, surpassing the effect of major world events such as the Covid pandemic.

source: <https://arxiv.org/abs/2406.07016>



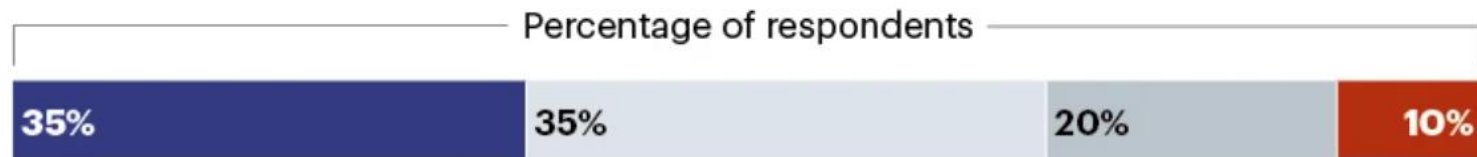


genAI in research

- Go to wooclap.com and use code RMAI1

Question: Dr Bloggs writes a first draft of their research article. Then they use ChatGPT (or another generative AI tool) to help edit the article, such as by asking for suggestions about its clarity, flow or tone. They don't disclose that they used AI. What is your view of this behaviour?

- Appropriate ■ Only appropriate if AI use disclosed
- Only appropriate if prompts disclosed
- Not appropriate under any circumstances



Question: Dr Bloggs uploads articles that they want to cite into an AI tool and asks for a short summary. They then use that AI-generated summary in their research paper, with the appropriate citation. They don't disclose they used AI. What is your view of this behaviour?

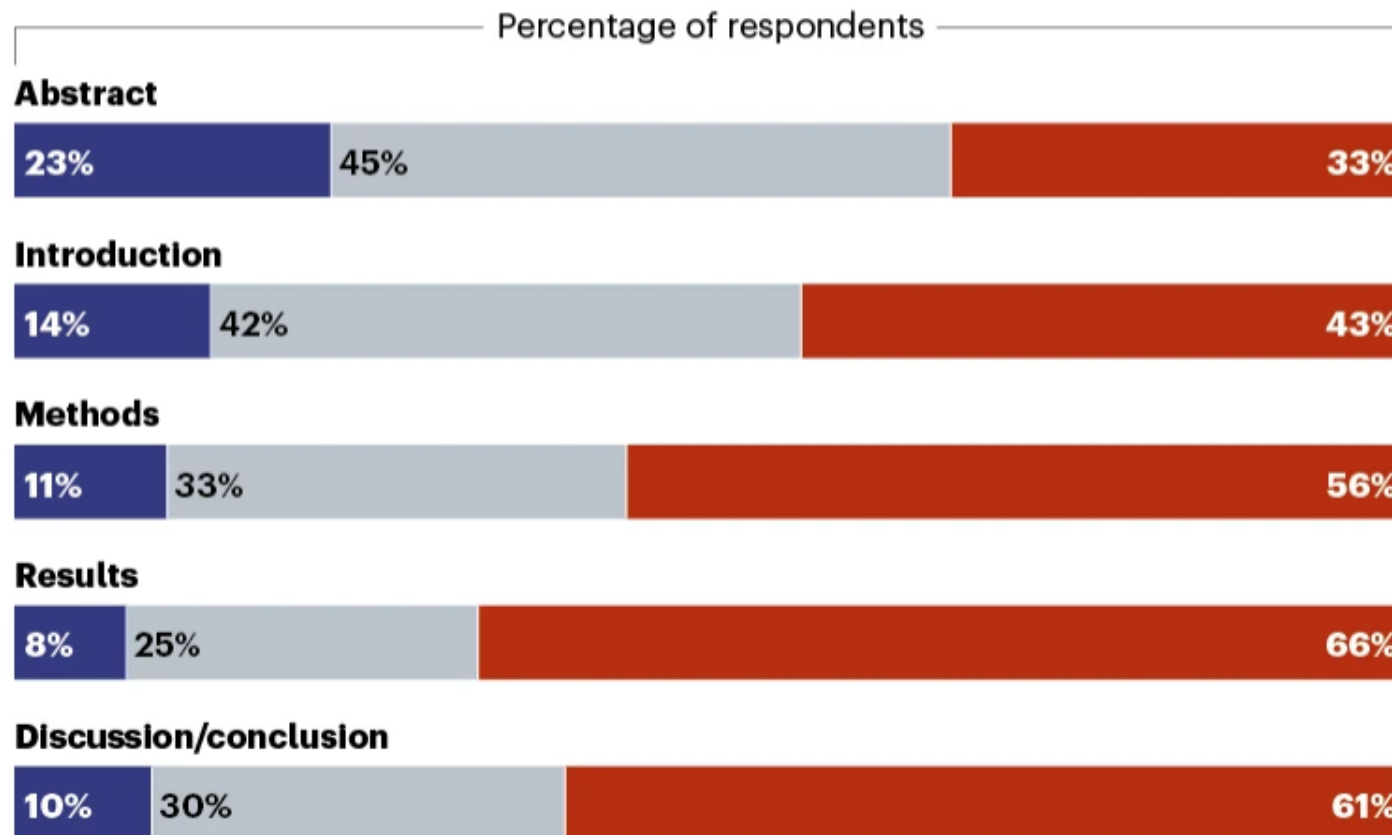
- Appropriate ■ Only appropriate if AI use disclosed
- Only appropriate if prompts disclosed
- Not appropriate under any circumstances



Nature's survey asked people to consider the following scenario: "Dr Bloggs uses an AI tool to write a section of a research paper, then edits the results. They don't disclose that they used AI." Respondents showed most concern about using AI for the results section.

Select your view of this behaviour for different sections of the paper

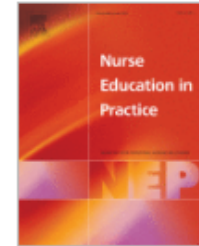
■ It's appropriate ■ Only appropriate with disclosure of AI use ■ Never appropriate





Nurse Education in Practice

Volume 66, January 2023, 103537



Editorial

Open artificial intelligence platforms in nursing education: Tools for academic progress or abuse?

Siobhan O'Connor^{a 1}  , ChatGPT^b 

source: <https://www.sciencedirect.com/science/article/abs/pii/S1471595322002517>



The first author became aware that the second listed author, 'ChatGPT', does not qualify for authorship according to the journal's guide for authors and to Elsevier's Publishing Ethics Policies.

'ChatGPT' is, therefore, removed from the author list and is acknowledged as making a substantial contribution to the writing of the paper. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

The journal and the author would like to apologize for any inconvenience caused.



genAI in research

- Policies may vary
- In general, not an author (no responsibility)
- Often: Statement on genAI use

Abbreviations

AI: Artificial Intelligence; RAG: Retrieval Augmented Generation; LLM: Large Language Model; HCPs: Health Care Professionals.

Supplementary Material

Supplementary appendix: question list.
<https://www.thno.org/v15p5693s1.pdf>

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Declaration of Generative AI usage

This manuscript utilized generative AI tools, including ChatGPT and Claude for linguistic improvement, DeepL for translation, and Grammarly for language correctness. Examples of prompts that were used for ChatGPT and Claude:

"Please review the following text for clarity and coherence: [Text]."

"Rewrite this paragraph to improve its flow and readability: [Text]."

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In this course

- genAI is not allowed unless for checking spelling, grammar, and other stylistic aspects
- Provide a detailed description of the used prompts and original output + a reflection on how it was used



In this course

- Always add one of these sentences (depending on which one applies)
 - “We did not use (generative) AI in the preparation and execution of this assignment beyond what is disclosed in the current section”
 - “No (generative) AI was used in the preparation and execution of this assignment.”



Overview of this lecture

- Scientific thinking
- The empirical cycle
- Validity
- genAI in research



Quiz

- A researcher states: “If this hypothesis is true, students using the app should score higher on the test.” Which phase is this?
 - ☐ A. Observation
 - ☐ B. Induction
 - ☐ C. Deduction
 - ☐ D. Evaluation

Quiz

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 - ☐ B. Induction
 - ☐ **C. Deduction**
 - ☐ D. Evaluation



Quiz

- What is internal validity mainly concerned with?
 - ☐ A. Whether the results can be generalized to other populations
 - ☐ B. Whether the study shows a trustworthy causal relationship
 - ☐ C. Whether participants enjoyed the study
 - ☐ D. Whether the sample size is large



Quiz

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 - ☐ **B. Whether the study shows a trustworthy causal relationship**
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